5G RAN

License Control Item Lists

Issue Draft B

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Security Declaration

Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:
https://www.huawei.com/en/psirt/vul-response-process
For vulnerability information, enterprise customers can visit the following web page:
https://securitybulletin.huawei.com/enterprise/en/security-advisory

Contents

1 License Control Item Lists

Licenses are contracts that specify the features, releases, capacities, validity periods, and usage scope of an operator's purchased products. For details about license principles, control methods, and operations related to license key files (LKFs), see *License Management*.

This document lists capacity-related license control items and feature-related license control items. The meaning of "Is Cell Activation Affected by License Insufficiency" mentioned in subsequent sections is as follows:

- Capacity licenses that affect cell activation: If the number of units of a license is insufficient during cell or hardware capacity expansion, the cell cannot be activated.
- Feature licenses that affect cell activation:
  - If a feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency.
  - If a feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation:
    - If the feature is enabled afterwards and the parameters that cause automatic cell reset are modified, the cell cannot be activated again due to license insufficiency.
    - If the feature supports the license sales number pre-check function (controlled by the **LIC_SALE_NUM_PRE_CHECK_SW** option of the **gNBLicenseAlmThld**.***LicenseChkSw*** parameter) and this function is enabled, the parameters for enabling the feature cannot be modified due to license insufficiency, preventing cell activation failures.
    - If the feature does not support the license sales number pre-check function, the parameters for enabling the feature cannot be modified due to license insufficiency, preventing cell activation failures.

  For details about the features involved in the preceding scenarios and the corresponding parameters for enabling these features, see section "Licenses" in the corresponding feature parameter description document.

1.1 Single-Mode Capacity-related License Control Items

A single-mode NR base station requires capacity-related license control items. If the single-mode NR base station uses 5000 series RF modules or UBBP boards, the corresponding First Mode License for 5000 Series RF Module(NR)/Massive MIMO Multimode License for 5000 Series RF Module (NR)/Multimode License for 5000 Series mmWave RF Module (NR TDD)/UBBP First-Mode License (NR) must also be purchased, in addition to the capacity-related license control items.

Table 1-1 describes the capacity items for NR. The **Model** and **Description** columns map the information in the **DSP LICINFO** command output. Note the following:

- If no LKF is available, the available capacity of a base station is the default capacity plus the capacity of the base station when no valid LKF is installed (default license state).
- If a valid LKF is installed, the available capacity of a base station equals the default capacity plus the capacity of the purchased license units.

Table 1-1 Capacity-related license control items

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S0BASIC10	gNodeB Basic Software, V100R010 (per Cell)	per Cell	**Function**: controls the number of physical cells that can be activated in a base station. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 1 license unit per base station **Default capacity**: none **Licensing principles**: One license unit is required for each activated physical cell. The number of license units to be purchased is equal to the number of physical cells that must be activated in a base station. **Total capacity**: Number of physical cells that can be activated in a base station = Number of purchased license units	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	Low-frequency TDD: Yes	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S0CBCBM00	10MHz Cell Bandwidth License for 5000 Series C-band RF Module (NR TDD)	per 10 MHz per Carrier	**Function**: controls the carrier bandwidth capability of 5000 series non-massive-MIMO C-band RF modules. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 10 license units per base station **Default capacity**: none **Licensing principles**: Carrier bandwidth is licensed in units of 10 MHz. One license unit is required for each 10 MHz carrier bandwidth on an RF module. The number of license units to be purchased is determined by the total number of 10 MHz carrier bandwidth units required by 5000 series non-massive-MIMO C-band RF modules in a base station. **Total capacity**: Carrier bandwidth available for 5000 series non-massive-MIMO C-band RF modules in a base station = Number of purchased license units x 10 MHz	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: No	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S0CB3GM00	5MHz Cell Bandwidth License for 5000 Series Sub-3 GHz RF Module (NR)	per 5MHz per Carrier	**Function**: controls the carrier bandwidth capability of 5000 series non-massive-MIMO sub-3 GHz RF modules. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 20 license units per base station **Default capacity**: none **Licensing principles**: Carrier bandwidth is licensed in units of 5 MHz. One license unit is required for each 5 MHz carrier bandwidth on an RF module. The number of license units to be purchased is determined by the total number of 5 MHz carrier bandwidth units required by 5000 series non-massive-MIMO sub-3 GHz RF modules in a base station. **Total capacity**: Carrier bandwidth available for 5000 series non-massive-MIMO sub-3 GHz RF modules in a base station = Number of purchased license units x 5 MHz	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: No	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S0CB3GM01	5MHz Cell Bandwidth License for Sub-3 GHz RF Module (NR)	per 5MHz per Carrier	**Function**: controls the carrier bandwidth capability of non-massive-MIMO sub-3 GHz RF modules. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 20 license units per base station **Default capacity**: none **Licensing principles**: Carrier bandwidth is licensed in units of 5 MHz. One license unit is required for each 5 MHz carrier bandwidth on an RF module. The number of license units to be purchased is determined by the total number of 5 MHz carrier bandwidth units required by non-massive-MIMO sub-3 GHz RF modules in a base station. **Total capacity**: Carrier bandwidth available for non-massive-MIMO sub-3 GHz RF modules in a base station = Number of purchased license units x 5 MHz	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: No	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S0PWRFM00	20W Power License for 5000 Series RF Module (NR)	per 20 W	**Function**: controls the maximum power of 5000 series non-massive-MIMO RF modules. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: none **Default capacity**: 20 W per RF module **Licensing principles**: RF power is licensed in units of 20 W. One license unit is required for every 20 W except for the default 20 W power per 5000 series non-massive-MIMO RF module. The number of license units to be purchased for a cell is determined by the cell power. **Total capacity**: Power available for 5000 series non-massive-MIMO RF modules in a base station = (Number of 5000 series non-massive-MIMO RF modules + Number of purchased license units) x 20 W	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: No	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S0PWRFM01	20W Power License for RF Module (NR)	per 20 W	**Function**: controls the maximum power of non-massive-MIMO RF modules. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: none **Default capacity**: 20 W per RF module **Licensing principles**: RF power is licensed in units of 20 W. One license unit is required for every 20 W except for the default 20 W power per non-massive-MIMO RF module. The number of license units to be purchased for a cell is determined by the cell power. **Total capacity**: Power available for non-massive-MIMO RF modules in a base station = (Number of non-massive-MIMO RF modules + Number of purchased license units) x 20 W	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: No	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S0MCRFM00	Multi-carrier License for RF Module (NR)	per Carrier	**Function**: controls the multi-carrier capability of non-massive-MIMO RF modules. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: none **Default capacity**: one carrier per RF module **Licensing principles**: One license unit is required for each carrier except for the default carrier per non-massive-MIMO RF module. **Total capacity**: Number of carriers available for a non-massive-MIMO RF module in a base station = Number of purchased license units + 1	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: No	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S0MCRFM01	Multi-carrier License for 5000 Series RF Module (NR)	per Carrier	**Function**: controls the multi-carrier capability of 5000 series non-massive-MIMO RF modules. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: none **Default capacity**: one carrier per RF module **Licensing principles**: One license unit is required for each carrier except for the default carrier per 5000 series non-massive-MIMO RF module. **Total capacity**: Number of carriers available for a 5000 series non-massive-MIMO RF module in a base station = Number of purchased license units + 1	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: No	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S0DLEPU00	Massive MIMO DL 2-Layers Extended Processing Unit License (NR)	per 2 Layers per Cell	**Function**: controls the capability of baseband processing units for transmitting different downlink layers through MIMO spatial multiplexing in massive-MIMO physical cells. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 2 license units per base station **Default capacity**: none **Licensing principles**: One license unit is required for every two downlink layers supported in each massive-MIMO physical cell served by a baseband processing unit. **Total capacity**: Number of downlink layers available for massive-MIMO physical cells on baseband processing units in a base station = Number of purchased license units x 2.	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: Yes	No

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S0ULEPU00	Massive MIMO UL 2-Layers Extended Processing Unit License (NR)	per 2 Layers per Cell	**Function**: controls the capability of baseband processing units for transmitting different uplink layers through MIMO spatial multiplexing in massive-MIMO physical cells. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 2 license units per base station **Default capacity**: none **Licensing principles**: One license unit is required for every two uplink layers supported in each massive-MIMO physical cell served by a baseband processing unit. **Total capacity**: Number of uplink layers available for massive-MIMO physical cells on baseband processing units in a base station = Number of purchased license units x 2.	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: Yes	No

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S000SPE00	RE(Resource Element) License (NR)	per RE	**Function**: controls the maximum number of users in RRC connected mode processing eMBB services supported by a base station. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 6 license units per base station **Default capacity**: none **Licensing principles**: One license unit is required for each RRC_CONNECTED UE performing eMBB services. **Total capacity**: Number of REs available in a base station = Number of purchased license units	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	Low-frequency TDD: Yes	No

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0SBBPABW00	UBBP Antenna Bandwidth 3*20 MHz* 2T2R License (NR)	per 3*20MHz*2T2R	**Function**: controls the antenna bandwidth capability of sub-6 GHz non-massive-MIMO FDD and TDD physical cells on a baseband processing unit. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 20 license units per base station **Default capacity**: none **Licensing principles**: One license unit is required for each 3 x 20 MHz x 2T2R antenna bandwidth on sub-6 GHz non-massive-MIMO physical cells served by a baseband processing unit. A cell bandwidth that is less than 20 MHz is considered to be 20 MHz bandwidth. A cell channel configuration lower than 2T2R is considered to be 2T2R. For example, three 100 MHz 8T8R non-massive-MIMO physical cells require 20 (= 5 x 4) license units. One 100 MHz 8T8R non-massive-	Low-frequency TDD	Macro gNodeB LampSite gNodeB	Low-frequency TDD: No	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
			MIMO physical cell requires seven (= Roundup (5 x 4)/3) license units. **Total capacity**: Antenna bandwidth available for sub-6 GHz non-massive-MIMO physical cells on the baseband processing units in a base station = Number of purchased license units x 3 x 20 MHz x 2T2R **NOTE** It is recommended that the **gNBLicenseConfig**.***AntBwLicensePolicy*** parameter be set to **COMM**.				

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0SBBPABW01	UBBP Antenna Bandwidth 3*20 MHz* 2T2R License (NR TDD)	per 3*20MHz*2T2R	**Function**: controls the antenna bandwidth capability of sub-6 GHz non-massive-MIMO TDD/SUL physical cells (with the subcarrier spacing of 30 kHz) on a baseband processing unit. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 20 license units per base station **Default capacity**: none **Licensing principles**: One license unit is required for each 3 x 20 MHz x 2T2R antenna bandwidth on sub-6 GHz non-massive-MIMO TDD/SUL physical cells (with the subcarrier spacing of 30 kHz) served by a baseband processing unit. A cell bandwidth that is less than 20 MHz is considered to be 20 MHz bandwidth. A cell channel configuration lower than 2T2R is considered to be 2T2R. Note: For a non-massive-MIMO TDD/SUL cell (with the	Low-frequency TDD	Macro gNodeB LampSite gNodeB	Low-frequency TDD: No	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
			subcarrier spacing of 30 kHz), the license control item NR0SBBPABW01 is preferentially used. If the license units are insufficient, NR0SBBPABW00 is used. When the license is deactivated, NR0SBBPABW00 is preferentially released. For example, three 100 MHz 8T8R non-massive-MIMO TDD physical cells require 20 (= 5 x 4) license units. One 100 MHz 8T8R non-massive-MIMO TDD physical cell requires seven (= Roundup (5 x 4)/3) license units. **Total capacity**: Antenna bandwidth available for sub-6 GHz non-massive-MIMO TDD/SUL physical cells (with the subcarrier spacing of 30 kHz) on the baseband processing units in a base station = Number of purchased license units x 3 x 20 MHz x 2T2R				

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
			NOTE It is recommended that the **gNBLicenseConfig**.***AntBwLicensePolicy*** parameter be set to **TDD_FDD**.				

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0SBPMMBW00	UBBP Massive MIMO Operating Bandwidth 1*20 MHz License (NR)	per 1*20MHz	**Function**: controls the bandwidth capability of sub-6 GHz massive-MIMO physical cells on a baseband processing unit. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 10 license units per base station **Default capacity**: none **Licensing principles**: One license unit is required for each 20 MHz bandwidth of a sub-6 GHz massive MIMO physical cell served by a baseband processing unit. A cell bandwidth less than 20 MHz is considered as 20 MHz. For example, one 64T64R 100 MHz sub-6 GHz massive MIMO physical cell requires 5 (100/20) license units. **Total capacity**: Bandwidth available for sub-6 GHz massive MIMO physical cells = Number of purchased license units x 1x20 MHz	Low-frequency TDD	Macro gNodeB LampSite gNodeB	Low-frequency TDD: Yes	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0SBPMMEC00	UBBP Massive MIMO Extended Channel 1*16T16R License (NR)	per 1*16T16R	**Function**: controls the channel capability of sub-6 GHz massive-MIMO physical cells on a baseband processing unit. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 3 license units per base station **Default capacity**: 16T16R per physical cell **Licensing principles**: One license unit is required for each 1x16T16R channels except for the default 16T16R channels per sub-6 GHz massive-MIMO physical cell. For example, one 64T64R sub-6 GHz massive-MIMO physical cell requires 3 license units. **Total capacity**: Number of channels available for sub-6 GHz massive-MIMO physical cells on the baseband processing units of a base station = Number of sub-6 GHz massive-MIMO physical cells x 16T16R	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: Yes	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
			+ Number of purchased license units x 1 x 16T16R				

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S0MMCBC00	Massive MIMO 10MHz Cell Bandwidth License for 5000 Series C-band RF Module (NR TDD)	per 10 MHz per Carrier	**Function**: controls the carrier bandwidth capability of 5000 series massive-MIMO C-band RF modules. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 10 license units per base station **Default capacity**: none **Licensing principles**: Carrier bandwidth is licensed in units of 10 MHz. One license unit is required for each 10 MHz carrier bandwidth on an RF module. The number of license units to be purchased is determined by the total number of 10 MHz carrier bandwidth units required by 5000 series massive-MIMO C-band RF modules in a base station. **Total capacity**: Carrier bandwidth available for 5000 series massive-MIMO C-band RF modules in a base station = Number of purchased license units x 10 MHz	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: Yes	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S0MMCBS00	Massive MIMO 5MHz Cell Bandwidth License for 5000 Series Sub-3 GHz RF Module (NR)	per 5MHz per Carrier	**Function**: controls the carrier bandwidth capability of 5000 series massive-MIMO sub-3 GHz RF modules. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 20 license units per base station **Default capacity**: none **Licensing principles**: Carrier bandwidth is licensed in units of 5 MHz. One license unit is required for each 5 MHz carrier bandwidth on an RF module. The number of license units to be purchased is determined by the total number of 5 MHz carrier bandwidth units required by 5000 series massive-MIMO sub-3 GHz RF modules in a base station. **Total capacity**: Carrier bandwidth available for 5000 series massive-MIMO sub-3 GHz RF modules in a base station = Number of purchased license units x 5 MHz	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: Yes	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S0MMPWL00	Massive MIMO 20W Power License for 5000 Series RF Module (NR)	per 20 W	**Function**: controls the maximum power of 5000 series massive-MIMO RF modules. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: none **Default capacity**: 20 W per RF module **Licensing principles**: RF power is licensed in units of 20 W. One license unit is required for every 20 W except for the default 20 W power per 5000 series massive-MIMO RF module. The number of license units to be purchased for a cell is determined by the cell power. **Total capacity**: Power available for 5000 series massive-MIMO RF modules in a base station = (Number of 5000 series massive-MIMO RF modules + Number of purchased license units) x 20 W	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: Yes	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S0MMMCL00	Massive MIMO Multi-carrier License for 5000 Series RF Module (NR)	per Carrier	**Function**: controls the multi-carrier capability of 5000 series massive-MIMO RF modules. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: none **Default capacity**: one carrier per RF module **Licensing principles**: One license unit is required for each carrier except for the default carrier per 5000 series massive MIMO RF module. **Total capacity**: Number of carriers available for a 5000 series massive-MIMO RF module in a base station = Number of purchased license units + 1	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: Yes	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S10MCBD00	10MHz Cell Bandwidth License for 5000 Series C-band LampSite pRRU(NR)	per 10 MHz per Carrier	**Function**: controls the carrier bandwidth capability of 5000 series C-band LampSite pRRUs. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 10 license units per base station **Default capacity**: none **Licensing principles**: Carrier bandwidth is licensed in units of 10 MHz. One license unit is required for each 10 MHz carrier bandwidth on a pRRU. The number of license units to be purchased is determined by the total number of 10 MHz carrier bandwidth units required by 5000 series C-band LampSite pRRUs in a base station. **Total capacity**: Carrier bandwidth available for 5000 series C-band LampSite pRRUs in a base station = Number of purchased license units x 10 MHz	Low-frequency TDD	LampSite gNodeB	Low-frequency TDD: No	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S5M3GBD00	5MHz Cell Bandwidth License for 5000 Series Sub-3G LampSite pRRU (NR)	per 5MHz per Carrier	**Function**: controls the carrier bandwidth capability of 5000 series sub-3 GHz LampSite pRRUs. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 20 license units per base station **Default capacity**: none **Licensing principles**: Carrier bandwidth is licensed in units of 5 MHz. One license unit is required for each 5 MHz carrier bandwidth on a pRRU. The number of license units to be purchased is determined by the total number of 5 MHz carrier bandwidth units required by 5000 series sub-3 GHz LampSite pRRUs in a base station. **Total capacity**: Carrier bandwidth available for 5000 series sub-3 GHz LampSite pRRUs in a base station = Number of purchased license units x 5 MHz	Low-frequency TDD	LampSite gNodeB	Low-frequency TDD: No	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S5POWER00	Power License for 5000 Series LampSite pRRU (NR)	per 125 mW	**Function**: controls the maximum power of 5000 series LampSite pRRUs. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: none **Default capacity**: 1 W per RF module **Licensing principles**: RF power is licensed in units of 125 mW. One license unit is required for every 125 mW except for the default 1 W power per 5000 series LampSite pRRU. The number of license units to be purchased for a cell is determined by the cell power. **Total capacity**: Power available for 5000 series LampSite pRRUs in a base station = Number of 5000 series LampSite pRRUs x 1 W + Number of purchased license units x 125 mW	Low-frequency TDD	LampSitegNodeB	Low-frequency TDD: No	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0SDMMIMO03	Distributed Massive MIMO for 5000 Series LampSite pRRU (NR)	per Carrier per pRRU	**Function**: controls the distributed massive MIMO capability of 5000 series LampSite pRRUs. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: none **Default capacity**: none **Licensing principles**: One license unit is required for each carrier on a 5000 series LampSite pRRU if distributed massive MIMO is enabled. **Total capacity**: Number of 5000 series LampSite pRRUs requiring distributed massive MIMO in a base station x Number of carriers per pRRU = Number of purchased license units	Low-frequency TDD	LampSitegNodeB	Low-frequency TDD: Yes	No

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S0RRCUL00	RRC Connected User License (NR)	per RRC Connected User	**Function**: controls the number of RRC_CONNECTED SA and NSA UEs on baseband processing units in a base station. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 6 license units per base station **Default capacity**: none **Licensing principles:** One license unit is required for each RRC_CONNECTED UE. The number of license units to be purchased is equal to the maximum number of RRC_CONNECTED UEs to be served by a base station. **Total capacity**: Maximum number of RRC_CONNECTED UEs to be served by a base station = Number of purchased license units	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	Low-frequency TDD: Yes	No

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0SMMWDLE00	mmWave DL 2-Layers Extended Processing Unit License (NR)	per 2 Layers per 100 MHz per Cell	**Function**: controls the capability of baseband processing units for transmitting different downlink layers through MIMO spatial multiplexing in mmWave physical cells. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 2 license units per base station **Default capacity**: none **Licensing principles**: One license unit is required for every two downlink layers per 100 MHz bandwidth supported in each mmWave physical cell served by a baseband processing unit. **Total capacity**: Number of downlink layers available for mmWave physical cells on baseband processing units in a base station = Number of purchased license units/(Cell bandwidth/100 MHz) x 2.	High-frequency TDD	Macro gNodeB LampSite gNodeB	-	No

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0SMMWULE00	mmWave UL 2-Layers Extended Processing Unit License (NR)	per 2 Layers per 100 MHz per Cell	**Function**: controls the capability of baseband processing units for transmitting different uplink layers through MIMO spatial multiplexing in mmWave physical cells. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 2 license units per base station **Default capacity**: none **Licensing principles**: One license unit is required for every two uplink layers per 100 MHz bandwidth supported in each mmWave physical cell served by a baseband processing unit. This license is not required when the mmWave physical cell is a supplementary cell. **Total capacity**: Number of uplink layers available for mmWave physical cells on baseband processing units in a base station = Number of purchased license units/(Cell	High-frequency TDD	Macro gNodeB LampSite gNodeB	-	No

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
			bandwidth/100 MHz) x 2.				
NR0SMMWBWT00	UBBP mmWave Operating Bandwidth 1*100 MHz License (NR)	per 1*100 MHz	**Function**: controls the bandwidth capability of mmWave physical cells on a baseband processing unit. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 8 license units per base station **Default capacity**: none **Licensing principles**: One license unit is required for each 100 MHz bandwidth of an mmWave physical cell served by a baseband processing unit. A cell bandwidth less than 100 MHz is considered as 100 MHz. For example, one mmWave 200 MHz physical cell requires two license units. **Total capacity**: Bandwidth available for mmWave physical cells served by a baseband processing unit = Number of purchased license units x 1 x 100 MHz	High-frequency TDD	Macro gNodeB LampSite gNodeB	-	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0SMMWECL00	UBBP mmWave Extended Channel 1*2T2R License (NR)	per 1*2T2R	**Function**: controls the channel capability of mmWave physical cells on a baseband processing unit. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 3 license units per base station **Default capacity**: 2T2R per physical cell **Licensing principles**: One license unit is required for each 1x2T2R channels except for the default 2T2R channels per mmWave physical cell. For example, one mmWave 4T4R physical cell requires one license unit. **Total capacity**: Number of channels available for mmWave physical cells on the baseband processing units of a base station = Number of mmWave physical cells x 2T2R + Number of purchased license units x 1 x 2T2R	High-frequency TDD	Macro gNodeB LampSite gNodeB	-	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S0CBMWM00	100MHz Cell Bandwidth License for 5000 Series mmWave RF Module (NR TDD)	per 100 MHz per Carrier	**Function**: controls the carrier bandwidth capability of 5000 series mmWave RF modules. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 4 license units per base station **Default capacity**: none **Licensing principles**: Carrier bandwidth is licensed in units of 100 MHz. One license unit is required for each 100 MHz carrier on an RF module. The number of license units to be purchased is determined by the total number of 100 MHz carrier bandwidth units required by 5000 series mmWave RF modules in a base station. **Total capacity**: Total carrier bandwidth available for 5000 series mmWave RF modules in a base station = Number of purchased license units x 100 MHz	High-frequency TDD	Macro gNodeB	-	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0SMCLMMW00	Multi-carrier License for 5000 Series mmWave RF Module (NR TDD)	per Carrier	**Function**: controls the multi-carrier capability of 5000 series mmWave RF modules. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: none **Default capacity**: one carrier per RF module **Licensing principles**: One license unit is required for each carrier except for the default carrier per 5000 series mmWave RF module. **Total capacity**: Number of carriers available for a 5000 series mmWave RF module in a base station = Number of purchased license units + 1	High-frequency TDD	Macro gNodeB	-	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0SEIRPMMW0	EIRP License for 5000 Series mmWave RF Module (NR TDD)	per 1dB	**Function**: controls the equivalent isotropically radiated power (EIRP) capability of 5000 series mmWave RF modules. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 30 license units per base station **Default capacity**: 56 dBm EIRP per RF module **Licensing principles**: EIRP is licensed in units of 1 dB. One license unit is required for each 1 dB EIRP except for the default 56 dBm EIRP per 5000 series mmWave RF module. The number of license units to be purchased for a cell is determined by the cell EIRP. **Total capacity**: EIRP available for 5000 series mmWave RF modules in a base station = Number of 5000 series mmWave RF modules x 56 dBm + Number of purchased license units x 1 dB	High-frequency TDD	Macro gNodeB	-	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0SMMWBAD00	100MHz Cell Bandwidth License for 5000 Series mmWave LampSite pRRU (NR TDD)	per 100 MHz per Carrier	**Function**: controls the carrier bandwidth capability of 5000 series mmWave RF modules. **Supported base stations**: DBS3900 and DBS5900 LampSite base stations **Capacity in the default license state**: none **Default capacity**: none **Licensing principles**: Carrier bandwidth is licensed in units of 100 MHz. One license unit is required for each 100 MHz carrier on an RF module. The number of license units to be purchased is determined by the total number of 100 MHz carrier bandwidth units required by 5000 series mmWave RF modules in a base station. A bandwidth less than 100 MHz is considered as 100 MHz bandwidth. **Total capacity**: (Bandwidth required by each carrier/100) x Accumulated number of pRRUs corresponding to each carrier = Number of purchased license units.	High-frequency TDD	LampSite gNodeB	-	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
			NOTE The value of (Bandwidth required by each carrier/ 100) is rounded up. For example, if the carrier bandwidth is 160 MHz, the value of (Bandwidth required by each carrier/ 100) is 2.				
NR0SMWMTUC00	Multi-carrier License for 5000 Series mmWave LampSite pRRU (NR TDD)	per Carrier	**Function**: controls the multi-carrier capability of 5000 series mmWave RF modules. **Supported base stations**: DBS3900 and DBS5900 LampSite base stations **Capacity in the default license state**: none **Default capacity**: one carrier per RF module **Licensing principles**: One license unit is required for each carrier except for the default carrier per 5000 series mmWave RF module. **Total capacity**: Number of carriers available for a 5000 series mmWave RF module in a base station = Number of purchased license units + Number of mmWave RF modules enabled with multi-carrier.	High-frequency TDD	LampSitegNodeB	-	Yes

Model	Description	Sales Unit	Details	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S0RCBWP00	RedCap BWP Hardware License (NR)	per BWP	**Function**: controls the number of RedCap BWPs that can be activated in a base station. **Supported base stations**: 3900 and 5900 series base stations **Capacity in the default license state**: 0 license unit per base station **Default capacity**: none **Licensing principles**: One license unit is required for each RedCap BWP in each physical cell. The number of license units to be purchased is equal to the number of RedCap BWPs that must be activated in each physical cell of a base station. **Total capacity**: Number of RedCap BWPs that need to be activated in all physical cells of a base station = Number of purchased license units	Low-frequency TDD	Macro gNodeB LampSite gNodeB	Low-frequency TDD: Yes	No

NOTE

In the preceding table, the symbol "-" means N/A.

NOTE

The usage of capacity resource items, such as the UE quantity, may significantly fluctuate during system operation. The usage is confined to the licensed limits even when it only slightly exceeds the licensed value. To prevent this problem and improve user experience, the usage of capacity resource items is allowed to reach 110% of the licensed value or the licensed value plus 10 (whichever value is larger).

1.2 Single-Mode Feature-related License Control Items

Feature items cover the following feature types:

- Basic software features

  Basic software features are the fundamental part of a software release.

  No separate license control item is available for each basic software feature. Instead, all basic software features are usually assembled into a mandatory basic software package for sale, such as the "gNodeB Basic Software" in **Table 1-1**. The ID of a basic software feature consists of four letters and six digits. For example, the ID of the basic feature Scheduling is FBFD-010011. For details about the basic feature list, see *Feature Navigation*.

- Optional software features

  Optional software features are licensed based on operators' requirements to provide values not enabled by basic software features. For details about the optional feature list, see *Feature Navigation*.

Table 1-2 Feature-related license control items

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00MUMM00	FOFD-010010	MU-MIMO Basic Pairing	per Cell	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	No	*MIMO (TDD)*

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S0PREUM00	FOFD-010020	SU-MIMO Multiple Layers	per Cell	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	No	*MIMO (TDD)*
NR0S0DUQAM00	FOFD-010050	DL 256QAM	per Cell	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	No	Modulation Schemes

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S0TNDER00	FOFD-010060	Transmission Network Detection and Reliability Improvement	per gNodeB	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	No	*IP Active Performance Measurement* *Ethernet OAM* *IPv4 Transmission* *IP NR Engineering Guide*
NR0S00STTN00	FOFD-010070	Network Synchronization	per gNodeB	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	No	Synchronization

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0SBSC3DC00	FOFD-010100	3D Coverage Pattern	per Cell	Low-frequency TDD	Macro gNodeB	Yes	Beam Management
NR0S00IBCA00	FOFD-020205	Intra-Band CA	per Cell	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	No	Carrier Aggregation
NR0S0GNBPS00	FOFD-021203	gNodeB Power Saving	per Cell	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	No	*Energy Conservation and Emission Reduction* *Green Site*

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S0AUTNE00	FOFD-021204	Automatic Neighbour Relation (ANR)	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	*ANR* *PCI Conflict Detection and Self-Optimization*

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0SRATMNE00	FOFD-021209	Inter-RAT Mobility from NG-RAN to E-UTRAN	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	*Interoperability Between E-UTRAN and NG-RAN in Idle Mode* *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00EPSF00	FOFD-021210	Voice Fallback	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Interoperability Between E-UTRAN and NG-RAN in Connected Mode
NR0S00MULS00	FOFD-021211	Multi-Operator Sharing	per Cell	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	Yes	Multi-Operator Sharing
NR0S000PV600	FOFD-021212	IPv6	per gNodeB	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	No	*IPv6 Transmission* *IP NR Engineering Guide*

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00UEPS00	FOFD-030210	UE Power Saving	per Cell	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	Yes	UE Power Saving
NR0S00SUPC00	FOFD-030211	Super Uplink	per Cell	Low-frequency TDD	Macro gNodeB	Yes	Super Uplink
NR0SSPUPPH01	FOFD-030211	Super Uplink	per Carrier per pRRU	Low-frequency TDD	LampSite gNodeB	Yes	Super Uplink
NR0SDORMAN10	FOFD-030216	pRRU Deep Dormancy	per pRRU	Low-frequency TDD	LampSite gNodeB	No	Multi-RAT Coordinated Energy Saving
NR0SINFRIBCA	FOFD-031201	Intra-FR Inter-Band CA	per Cell	Low-frequency TDD	Macro gNodeB	No	Carrier Aggregation
NR0TSIFIBC01	FOFD-031201	Intra-FR Inter-Band CA	per Carrier per pRRU	Low-frequency TDD	LampSite gNodeB	No	Carrier Aggregation

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S000PSS00	FOFD-031204	Intelligent Scheduling for Power Saving	per Cell	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	No	Energy Conservation and Emission Reduction
NR0S00RCIS00	FOFD-031205	RF Channel Intelligent Shutdown	per Cell	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	No	Energy Conservation and Emission Reduction
NR0S00ICSC00	FOFD-050203	Intelligent Carrier Shutdown	per Cell	Low-frequency TDD High-frequency TDD	Macro gNodeB	No	Energy Conservation and Emission Reduction

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0SSHUTDW00	FOFD-050203	Intelligent Carrier Shutdown	per Carrier per pRRU	Low-frequency TDD	LampSite gNodeB	No	Energy Conservation and Emission Reduction
NR0S00HPCL00	FOFD-030205	Hyper Cell	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	Yes	Hyper Cell
NR0SDMMIMO00	FOFD-050202	Distributed Massive MIMO	per Carrier per pRRU	Low-frequency TDD	LampSite gNodeB	Yes	Distributed Antenna Solutions (Low-Frequency TDD)
NR0S0NRSFN00	FOFD-031207	Cell Combination	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	Yes	Cell Combination

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S0IPSEC00	FOFD-010080	IPsec	per gNodeB	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	No	*IPsec* *PKI* *eXn Self-Management* *NG and Xn Self-Management*
NR0S0FWAPL00	FOFD-051205	FWA Private Line	per Cell	Low-frequency TDD High-frequency TDD	Macro gNodeB	No	FWA
NR0SULADLD00	FOFD-010205	UL and DL Decoupling	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	Yes	UL and DL Decoupling

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0SMMW3DCP0	FOFD-030201	mmWave 3D Coverage Pattern	per Cell	High-frequency TDD	Macro gNodeB LampSite gNodeB	Yes	mmWave Beam Management (High-Frequency TDD)
NR0S000CMP00	FOFD-030204	CoMP	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	CoMP
NR0S00HSMB00	FOFD-030207	High-speed Railway Superior Experience	per Cell	Low-frequency TDD	Macro gNodeB	Yes	High Speed Mobility

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00RIMG00	FOFD-030208	Remote Interference Management (RIM)	per Cell	Low-frequency TDD	Macro gNodeB	No	Remote Interference Management (Low-Frequency TDD)
NR0S00CAWW00	FOFD-050206	CA SRS Carrier Switching	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	Yes	CA Experience Improvement
NR0S00EXCR00	FOFD-051201	Extended Cell Range	per Cell	Low-frequency TDD	Macro gNodeB	Yes	Extended Cell Range
NR0S00IPCA00	FOFD-051202	Inter-gNodeB CA	per Cell	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	No	CA Experience Improvement

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00VONR00	FOFD-031203	VoNR	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	*VoNR* *ViNR*
NR0S00EFWA00	FOFD-051204	Experience-based FWA	per Cell	Low-frequency TDD High-frequency TDD	Macro gNodeB	No	FWA
NR0S00PSCH00	FOFD-050208	Service-Differentiated Low Latency	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Service-differentiated Experience Guarantee
NR0S00BNSL00	FOFD-050201	Network Slicing Introduction	per Cell	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	No	Network Slicing
NR0SMATPLS00	FOFD-050207	Precise Positioning (LampSite NR) (Trial)	per Cell	Low-frequency TDD	LampSite gNodeB	No	Positioning

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00CDIM00	FOFD-051210	Coordinated Interference Management	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Coordinated Interference Management (Low-Frequency TDD)
NR0SUL256QAM	FOFD-030215	UL 256QAM	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Modulation Schemes

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S000CLB00	FOFD-051212	Converged Load Balancing	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	*Mobility Load Balancing in Connected Mode* *Mobility Load Balancing in Idle Mode*
NR0S00AET100	FOFD-051301	AHR Introduction	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	Yes	Massive MIMO AHR (Low-Frequency TDD)

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00VGMC00	FOFD-060201	Virtual Grid-based Multi-Frequency Coordination	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Virtual Grid-based Multi-Frequency Coordination
NR0SESBFFS00	FOFD-060203	Energy Saving Based on Flexible Frequency-Domain Scheduling	per Cell	Low-frequency TDD	Macro gNodeB	No	Energy Conservation and Emission Reduction
NR0S03DNEM00	FOFD-060202	3D Networking Experience Improvement	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	Yes	3D Networking Experience Improvement

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00VNPE00	FOFD-060301	VoNR Performance Enhancement	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	VoNR
NR0S00UCVE00	FOFD-081207	Deterministic Latency Guarantee for CloudX	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Ultimate Cloud X experience
NR0S0FASTCA0	FOFD-061224	Fast CA	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	CA Experience Improvement
NR0S0SCSNR00	FOFD-061223	Experience Boosting based on Multi-Band Coordination	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Multi-Frequency Smart Selection

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00ATFS00	FOFD-061220	Adaptive Interference Suppression	per Cell	Low-frequency TDD	Macro gNodeB	No	Remote Interference Management (Low-Frequency TDD)
NR0S00TBUL00	FOFD-061205	Turbo Uplink	per Cell	Low-frequency TDD	Macro gNodeB	No	Turbo Uplink (Low-Frequency TDD)

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00CDPS00	FOFD-061222	NR Inter-Carrier Dynamic Power Sharing	per Cell	Low-frequency TDD	Macro gNodeB	No	NR Inter-Carrier Dynamic Power Allocation
NR0S00AET200	FOFD-061201	AHR Experience Turbo 2.0	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Massive MIMO AHR (Low-Frequency TDD)
NR0S00ACT200	FOFD-061202	AHR Capacity Upgrade 2.0	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Massive MIMO AHR (Low-Frequency TDD)

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00SFBS00	FOFD-070201	Multi-Frequency Spatial Smart Carrier Selection	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	*Multi-Frequency Smart Selection* *Multi-Frequency Smart Aggregation*
NR0S00COMM00	FOFD-070203	Coordinated Massive MIMO (Co-MM)	per Cell	Low-frequency TDD	Macro gNodeB	No	Co-MM (Low-Frequency TDD)

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00PARI00	FOFD-061292	Proactive Avoidance of Remote Interference	per Cell	Low-frequency TDD	Macro gNodeB	Yes	Remote Interference Management (Low-Frequency TDD)
NR0S00MMCE00	FOFD-070301	mmWave Coverage Extension	per Cell	High-frequency TDD	Macro gNodeB	Yes	FWA
NR0S00RCID00	FOFD-070316	RedCap Introduction	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	RedCap
NR0S00RCCE00	FOFD-091206	Premium RedCap	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	RedCap
NR0S00HPPU00	FOFD-070210	Precise Positioning (NR) (Trial)	per Cell	Low-frequency TDD	Macro gNodeB	No	Positioning

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00DRCS00	FOFD-061203	Intelligent Channel Dynamic Shutdown	per Cell	Low-frequency TDD	Macro gNodeB	No	Energy Conservation and Emission Reduction
NR0S0PSBPC00	FOFD-071903	Intelligent Power Control Energy Saving	per Cell	Low-frequency TDD	Macro gNodeB	No	Energy Conservation and Emission Reduction
NR0SMRMCES00	FOFD-071904	Intelligent Multi-RF-Module Coordinated Energy Saving	per Cell	Low-frequency TDD	Macro gNodeB	No	Energy Conservation and Emission Reduction

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S0MMSCS00	FOFD-071901	Macro-Micro Smart Carrier Selection	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Multi-Frequency Smart Selection
NR0SURLLCP00	FOFD-051206	URLLC Introduction (Trial)	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	URLLC
NR0S00AML000	FOFD-071203	Adaptive Managed Latency	per gNodeB	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	*Adaptive Managed Latency* *EPC-based NSA Basics*
NR0S00KQAM00	FOFD-081237	DL 1024QAM	per Cell	Low-frequency TDD	Macro gNodeB	No	Modulation Schemes

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00MRPT00	FOFD-070320	Mobility Robustness Optimization	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	MRO
NR0S005LAN00	FOFD-050209	5G LAN Transport	per Cell	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	No	5G LAN Transport
NR0SDL3CCA00	FOFD-051211	3CC CA	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	CA Experience Improvement
NR0S00TBUL20	FOFD-081202	Turbo Uplink 2.0	per Cell	Low-frequency TDD	Macro gNodeB	No	Turbo Uplink (Low-Frequency TDD)

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00VMMM00	FOFD-071211	Distributed Massive MIMO (NR TDD)	per Cell	Low-frequency TDD	Macro gNodeB	Yes	Distributed Antenna Solutions (Low-Frequency TDD)
NR0S00MCEA00	FOFD-081205	CA Performance Boost Based on Multi-dimensional Optimization	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Multi-Frequency Smart Aggregation
NR0S00SVSE00	FOFD-080011	Stable Video Surveillance Experience	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Stable Video Surveillance Experience

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00MMCA00	FOFD-080012	Massive CA	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	CA Experience Improvement
NR0S00NUEG00	FOFD-080019	Intelligent Uplink Experience Guarantee (Trial)	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Ultimate Video Experience
NR0S08LSFU00	FOFD-081231	8 Layers for a Single FWA User (Trial)	per Cell	Low-frequency TDD	Macro gNodeB	No	FWA
NR0S0DPSFSA0	FOFD-081220	FWA Service Turbo	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	FWA
NR0S00BEAM00	FOFD-081201	iBeam	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	Yes	Massive MIMO iBeam (Low-Frequency TDD)

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00MTDD00	FOFD-081203	MetaAAU Deep Energy Saving	per Cell	Low-frequency TDD	Macro gNodeB	No	Energy Conservation and Emission Reduction
NR0S00RARA00	FOFD-081208	Inter-Mode Interference Suppression	per Cell	Low-frequency TDD	Macro gNodeB	No	Interference Avoidance
NR0S00UPSB00	FOFD-081252	UE Power Saving Based on Inactive State	per Cell	Low-frequency TDD	Macro gNodeB	No	UE Power Saving

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00PCDW00	FOFD-101241	Uplink IAS	per Cell	High-frequency TDD	Macro gNodeB	No	Intelligent Aggregate Scheduling (High-Frequency TDD)
NR0S00ULUL00	FOFD-101240	Ultra Uplink	per Cell	High-frequency TDD	Macro gNodeB	No	Ultra Uplink (High-Frequency TDD)

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00MWBM00	FOFD-101242	mmWave Super Coverage	per Cell	High-frequency TDD	Macro gNodeB	No	mmWave Super Coverage (High-Frequency TDD)
NR0S00FSSD00	FOFD-081250	Fast Carrier Shutdown	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Energy Conservation and Emission Reduction
NR0S00FRCA00	FOFD-070206	Inter-FR CA (Trial)	per Cell	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	No	Carrier Aggregation

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S000SDC00	FOFD-081234	Single DCI (Trial)	per Cell	Low-frequency TDD	Macro gNodeB	No	Multi-Frequency Smart Aggregation
NR0S00UMFS00	FOFD-081260	Uplink Multi-band Flexible Scheduling (Trial)	per Cell	Low-frequency TDD	Macro gNodeB	No	Uplink Flexible Spectrum Scheduling
NR0S00RDEE00	FOFD-091204	RedCap Downlink Experience Improvement (NR TDD)	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	Yes	RedCap

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00UAHR00	FOFD-091201	Uplink Boosting	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Massive MIMO Uplink Boosting (Low-Frequency TDD)
NR0S00MBPA00	FOFD-090100	Multi-band Power Adaptive Scheduling	per Cell	Low-frequency TDD	Macro gNodeB	No	*LTE and NR Power Allocation* *NR Inter-Carrier Dynamic Power Allocation*

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00ULSG00	FOFD-091253	Uplink Service Guarantee	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Ultimate Video Experience
NR0S0DLEHR00	FOFD-091200	iBeam 2.0	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Massive MIMO iBeam (Low-Frequency TDD)
NR0S00DSCS00	FOFD-091209	Latency-based Optimal Carrier Selection (NR)	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Multi-Frequency Smart Selection
NR0S00LQS000	FOFD-091203	Layered QoS	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Layered QoS

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S00LSG000	FOFD-091202	Game-like Service Latency Assurance	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Service-differentiated Experience Guarantee
NR0S00QSCP10	FOFD-091230	QoS-based CA Policy	per gNodeB	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Service-differentiated Experience Guarantee
NR0S00UPBT20	FOFD-100201	Uplink Boosting 2.0	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Massive MIMO Uplink Boosting (Low-Frequency TDD)

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0SPCPWMG00	FOFD-100252	AMC-based Precise Power Control Energy Saving	per Cell	Low-frequency TDD	Macro gNodeB	No	Energy Conservation and Emission Reduction
NR0S00FULC30	FOFD-100206	FWA Ultra-Large Coverage	per Cell	Low-frequency TDD	Macro gNodeB	No	FWA
NR0S00BEAM30	FOFD-100200	iBeam 3.0	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Massive MIMO iBeam (Low-Frequency TDD)
NR0S00ULSF40	FOFD-100205	Uplink 4 Layers for a Single FWA User	per Cell	Low-frequency TDD	Macro gNodeB	No	FWA

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0SFLEXGR10	FOFD-100210	Flexible Rate Grading (trial)	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	*Service-differentiated Experience Guarantee*
NR0SNTBACD00	FOFD-101225	L/NR Inter-band Deep Coordination	per Cell	Low-frequency TDD	Macro gNodeB	No	LTE and NR Power Allocation

NOTE

In a hyper cell or combined cell, the number of required license units to enable a feature whose license sales unit is "per Cell" is calculated based on each physical cell in the hyper cell or combined cell, instead of each hyper cell or combined cell.

For example, if three physical cells form a hyper cell or combined cell and DL 256QAM is enabled in the hyper cell or combined cell, then three license units for Hyper Cell or Cell Combination and three license units for DL 256QAM are required.

1.3 Multimode Capacity-related License Control Items

Multimode items apply to the RATs in multimode modules, and items of this type can only be used after they have been purchased for the corresponding RAT. Currently, multimode capacity-related license control items are classified into four type: RF multimode, RF spectrum sharing, UBBP multimode, and UMPT multimode. **Table 1-3** summarizes these license control items.

Table 1-3 Multimode capacity-related license control items

Model	Description	Sales Unit	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S00UMMG00	UBBP First-Mode license (NR)	per UBBP	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	Low-frequency TDD: Yes	Yes
NR0S00UMMG01	UBBP Multi-Mode license (NR)	per UBBP	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	Low-frequency TDD: Yes	Yes
NR0SUMPTMM00	UMPT Multi Mode license (NR)	per UMPT	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	Low-frequency TDD: Yes	Yes
NR0S5000MMRR	Multimode License for 5000 Series RF Module (NR)	per RU	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: No	Yes
NR0S5000FMRR	First Mode License for 5000 Series RF Module (NR)	per RU	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: No	Yes

Model	Description	Sales Unit	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0S00RFDM00	Multimode License for RF Module (NR)	per Sector	Low-frequency TDD High-frequency TDD	Macro gNodeB	Low-frequency TDD: No	Yes
NR0SMBRFMD00	Multi-band License for 5000 Series Multi-band RF Module (NR)	per Band per RU	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: No	Yes
NR0SMMRFMD00	Massive MIMO Multimode License for 5000 Series RF Module (NR)	per RU	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: Yes	Yes
NR0SMMMBMD00	Massive MIMO Multi-band License for 5000 Series Multi-band RF Module (NR)	per Band per RU	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: Yes	Yes
NR0SFMFDMD00	First Mode First Band License for Multi-Band RF Module (NR)	per Band per RU	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: No	Yes
NR0SMMMBMB00	Multi-Mode Multi-Band License for Multi-Band RF Module (NR)	per Band per RU	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: No	Yes
NR0S0RFSSL00	RF Spectrum Sharing License (NR)	per Band per RU	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: No	Yes

Model	Description	Sales Unit	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0SSPECTT00	Spectrum Sharing License for LampSite pRRU (NR)	per Band per pRRU	Low-frequency TDD	LampSite gNodeB	Low-frequency TDD: No	Yes
NR0SSSLRFM00	Spectrum Sharing License for 5000 Series RF Module (NR)	per Band per RU	Low-frequency TDD	Macro gNodeB	Low-frequency TDD: No	Yes
NR0SSPECTT50	Spectrum Sharing License for 5000 Series LampSite pRRU (NR)	per Band per pRRU	Low-frequency TDD	LampSite gNodeB	Low-frequency TDD: No	Yes
NR0SFSTMOD00	First Mode License for 5000 Series LampSite pRRU(NR)	per pRRU	Low-frequency TDD	LampSite gNodeB	Low-frequency TDD: No	Yes
NR0SMUTMOD00	Multimode License for 5000 Series LampSite pRRU (NR)	per pRRU	Low-frequency TDD	LampSite gNodeB	Low-frequency TDD: No	Yes
NR0SMMWAMD00	Multimode License for 5000 Series mmWave RF Module (NR TDD)	per RU	High-frequency TDD	Macro gNodeB	-	Yes
NR0SFMFBMBL00	LampSite First Mode First Band License for Multi-Band RF Module(NR)	per Band per pRRU	Low-frequency TDD	LampSite gNodeB	Low-frequency TDD: No	Yes

Model	Description	Sales Unit	RAT	NE	Is Massive MIMO Involved	Is Cell Activation Affected by License Insufficiency
NR0SMMMBMBL00	LampSite Multi-Mode Multi-Band License for Multi-Band RF Module (NR)	per Band per pRRU	Low-frequency TDD	LampSite gNodeB	Low-frequency TDD: No	Yes
NR0SMWMTUM00	Multimode License for 5000 Series mmWave LampSite pRRU (NR TDD)	per pRRU	High-frequency TDD	LampSite gNodeB	-	Yes

RF multimode license control items: Different types of RF modules use different multimode license solutions. If a base station uses different types of RF modules, different multimode license solutions must be used to sell and control these RF modules.

- When a single-band non-5000 series RF module is configured to serve two RATs (GU/GL/UL), the operator needs to purchase the corresponding RF module dual-mode (GU/GL/UL) license. When such a single-band non-5000 series RF module is configured to serve two RATs (LN), the operator needs to purchase Multimode License for RF Module (NR). The RF module dual-mode (GU/GL/UL) license is sold and controlled based on the more recent RAT of the two RATs (LTE TDD > LTE FDD > UMTS > GSM). When the RF module serves a single RAT (GO/UO/LO/NO), the operator does not need to purchase RF module multimode licenses.
- Single-band 5000 series RFUs, RRUs, and AAUs can share the same type of license. The differences are detailed as follows:

  For sub-6 GHz RF modules of 8T8R or lower, if such an RF module is working only for one RAT, the operator needs to purchase a First Mode License for 5000 Series RF Module for this RAT. If the RF module is serving multiple RATs, each RAT requires one license. Specifically, the lowest RAT uses a First Mode License for 5000 Series RF Module while other RATs use Multimode License for 5000 Series RF Module.

  For sub-6 GHz RF modules of 16T16R or higher, each RAT requires a Massive MIMO Multimode License for 5000 Series RF Module.

  High-frequency RF modules apply only to NR. Each high-frequency RF module requires one Multimode License for 5000 Series mmWave RF Module (NR TDD).

- For sales of multiband non-5000 series RF modules, RF module licenses are classified into the First Mode First Band License for Multi-Band RF Module and the Multi-Mode Multi-Band License for Multi-Band RF Module for each RAT (GSM, UMTS, LTE, and NR). The First Mode First Band License for Multi-Band RF Module is automatically converted into the Multi-Mode Multi-Band License for Multi-Band RF Module during usage. Therefore, for a multiband RF module, one Multi-Mode Multi-Band License for Multi-Band RF Module is required for each frequency band providing GSM, UMTS, LTE, or NR services.
- For multiband 5000 series RF modules, the differences are detailed as follows:

  For sub-6 GHz RF modules of 8T8R or lower, if more than one frequency band is used for NR, each additional frequency band requires one Multi-band License for 5000 Series Multi-band RF Module.

  For sub-6 GHz RF modules of 16T16R or higher, if more than one frequency band is used for NR, each additional frequency band requires one Massive MIMO Multi-band License for 5000 Series Multi-band RF Module.
- If a LampSite 5000 series multi-band pRRU serves only one RAT, the operator needs to purchase a First Mode License for 5000 Series LampSite pRRU for this RAT. If the pRRU serves multiple RATs, each RAT requires one license. Specifically, the lowest RAT uses a First Mode License for 5000 Series LampSite pRRU while other RATs use Multimode License for 5000 Series LampSite pRRU.

Table 1-4 lists RF multimode license control items.

Table 1-4 RF multimode license control items

NE Type	Model	Description
gNodeB	NR0S5000FMRR	First Mode License for 5000 Series RF Module (NR)
gNodeB	NR0SMWMTUM00	mmWave Multimode License for 5000 Series LampSite pRRU (NR)
gNodeB	NR0S5000MMRR	Multimode License for 5000 Series RF Module (NR)
gNodeB	NR0S00RFDM00	Multimode License for RF Module (NR)
gNodeB	NR0SMBRFMD00	Multi-band License for 5000 Series Multi-band RF Module (NR)
gNodeB	NR0SMMRFMD00	Massive MIMO Multimode License for 5000 Series RF Module (NR)
gNodeB	NR0SMMMBMD00	Massive MIMO Multi-band License for 5000 Series Multi-band RF Module (NR)
gNodeB	NR0SMMWAMD00	Multimode License for 5000 Series mmWave RF Module (NR TDD)
gNodeB	NR0SFMFDMD00	First Mode First Band License for Multi-Band RF Module (NR)

NE Type	Model	Description
gNodeB	NR0SMMMBMB00	Multi-Mode Multi-Band License for Multi-Band RF Module (NR)
gNodeB	NR0SFSTMOD00	First Mode License for 5000 Series LampSite pRRU(NR)
gNodeB	NR0SMUTMOD00	Multimode License for 5000 Series LampSite pRRU (NR)
gNodeB	NR0SFMFBMBL00	LampSite First Mode First Band License for Multi-Band RF Module(NR)
gNodeB	NR0SMMMBMBL00	LampSite Multi-Mode Multi-Band License for Multi-Band RF Module (NR)

Spectrum sharing license: For an RF module, if spectrum sharing-related features are enabled for the same RAT on different bands, the number of spectrum sharing licenses required equals the number of frequency bands; if multiple spectrum sharing-related features are enabled across multiple frequencies for one RAT on one band, only one spectrum sharing license is required. Non-5000 series RF modules and 5000 series RF modules consume different RF spectrum sharing licenses, and RF modules of different RATs consume different RF spectrum sharing licenses. The following scenarios take 5000 series RF modules as an example:

- Scenario 1: If LTE cells enabled with MRFD-131223 LTE FDD and NR Uplink Spectrum Sharing (LTE FDD) are configured on a 5000 series RF module, the following license item is required: Spectrum Sharing License for 5000 Series RF Module (FDD).
- Scenario 2: If NR cells enabled with MRFD-131263 LTE FDD and NR Uplink Spectrum Sharing (NR) are configured on a 5000 series RF module, the following license item is required: Spectrum Sharing License for 5000 Series RF Module (NR).
- Scenario 3: If LTE cells enabled with MRFD-131223 LTE FDD and NR Uplink Spectrum Sharing (LTE FDD) and NR cells enabled with MRFD-131263 LTE FDD and NR Uplink Spectrum Sharing (NR) are configured on a 5000 series RF module, the following license items are required: Spectrum Sharing License for 5000 Series RF Module (FDD) and Spectrum Sharing License for 5000 Series RF Module (NR).

NOTE

Spectrum sharing features include:

- MRFD-131223 LTE FDD and NR Uplink Spectrum Sharing (LTE FDD) for LTE cells
- MRFD-131263 LTE FDD and NR Uplink Spectrum Sharing (NR) for NR cells

Table 1-5 lists RF spectrum sharing license control items.

Table 1-5 RF spectrum sharing license control items

NE Type	Model	Description
Macro eNodeB	LT1SRFSPCS00	RF Spectrum Sharing License (FDD)
Macro eNodeB	LT4S5RFSSTDD	Spectrum Sharing License for 5000 Series RF Module(TDD)
Macro eNodeB	LT1S5000RFSS	Spectrum Sharing License for 5000 Series RF Module (FDD)
Macro gNodeB	NR0S0RFSSL00	RF Spectrum Sharing License (NR)
Macro gNodeB	NR0SSSLRFM00	Spectrum Sharing License for 5000 Series RF Module (NR)
LampSite eNodeB	LT4SSPECTT00	Spectrum Sharing License for 5000 Series LampSite pRRU (TDD)
LampSite gNodeB	NR0SSPECTT50	Spectrum Sharing License for 5000 Series LampSite pRRU (NR)
LampSite eNodeB	LT1SFDSPCS30	Spectrum Sharing License for LampSite pRRU (LTE FDD)
LampSite eNodeB	LT1SFDSPCS50	Spectrum Sharing License for 5000 Series LampSite pRRU(LTE FDD)
LampSite gNodeB	NR0SSPECTT00	Spectrum Sharing License for LampSite pRRU (NR)

UBBP multimode licenses: The UBBP is a multimode baseband processing unit that can also be used for a single RAT and requires the license for at least one RAT. For NR or LTE, UBBP multimode licenses are classified into the UBBP First-Mode License and UBBP Multi-Mode License. When a UBBP works for a single RAT, the operator only needs to purchase the UBBP First-Mode License for this RAT. However, the licenses for each added RAT are required for base station RAT extension. When deploying a new multimode base station, the operator needs to purchase one UBBP First-Mode License and *N*-1 UBBP Multi-Mode Licenses (*N* indicates the total number of RATs).

Table 1-6 lists UBBP multimode license control items.

Table 1-6 UBBP multimode license control items

NE Type	Model	Description
eNodeB	WDMS0UFMLF00	UBBP First-Mode License (LTE FDD)
eNodeB	WDMS0UMMLF00	UBBP Multi-Mode License (LTE FDD)
eNodeB	LT1SUMMLFL01	UBBP First-Mode License (LTE TDD)

NE Type	Model	Description
eNodeB	LT1SUMMLFL00	UBBP Multi-Mode License (LTE TDD)
eNodeB	ML1SUBBPFM00	UBBP First-Mode License (NB-IoT)
eNodeB	ML1SUBBPMM00	UBBP Multi-Mode License (NB-IoT)
gNodeB	NR0S00UMMG00	UBBP First-Mode license (NR)
gNodeB	NR0S00UMMG01	UBBP Multi-Mode license (NR)

UMPT multimode license: There is no need to purchase a UMPT multimode license when a UMPT works in a single RAT. However, the licenses for each added RAT are required for base station RAT extension. When deploying a new multimode base station, the operator needs to purchase *N*–1 license units for *N* RATs (*N* indicates the total number of RATs in a multimode base station). For example, if a UMPT works in LTE and NR, the operator needs to purchase an LTE or NR license. If the licensed value and usage value of this license control item queried on the gNodeB are **0** and **1**, respectively, and ALM-26819 Data Configuration Exceeding Licensed Limit is not reported, the usage value of **1** indicates the default authorized value when no license is available. This is a normal symptom that does not need to be handled.

Table 1-7 lists UMPT multimode license control items.

Table 1-7 UMPT multimode license control items

NE Type	Model	Description
eNodeB	WDMS0UMML00	UMPT Multi Mode license(LTE FDD)
eNodeB	LT1UMPTMMS00	UMPT Multi Mode license(LTE TDD)
eNodeB	ML1SUMPTMM00	UMPT Multi Mode license(NB-IoT)
gNodeB	NR0SUMPTMM00	UMPT Multi Mode license (NR)

NOTE

The current license control does not distinguish between First Mode License for 5000 Series RF Module (NR) and Multimode License for 5000 Series RF Module (NR). If both license control items are purchased, you can run the **DSP LICINFO** command to query license information. In the command output, the **Allocated** field indicates the licensed values of the two control items, the **Config** field indicates the sum of the two control items' configured values, and the **Actual Used** field indicates the sum of the two control items' actually used values. Therefore, the values displayed in columns **Config** and **Actual Used** in the command output for one license control item are the same as those for the other. The license information for the following pairs of license control items can be queried in the **DSP LICINFO** command output in the same way: First Mode First Band License for Multi-Band RF Module (NR) and Multi-Mode Multi-Band License for Multi-Band RF Module (NR), UBBP First-Mode License (NR) and UBBP Multi-Mode license (NR), and First Mode License for 5000 Series LampSite pRRU(NR) and Multimode License for 5000 Series LampSite pRRU(NR).

1.4 Multimode Feature-related License Control Items

Table 1-8 Multimode feature-related license control items

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0SENDCCE00	MRFD-151263	EN-DC Performance Enhancement (NR)	per Cell	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	No	EPC-based NSA Performance Enhancement
NR0S0LNICS00	MRFD-160263	LTE and NR Intelligent Carrier Shutdown (NR)	per Cell	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	Multi-RAT Coordinated Energy Saving

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0SMRCSPS00	MRFD-160266	Multi-RAT Coordinated Symbol Power Saving (NR)	per Cell	Low-frequency TDD	Macro gNodeB	No	Multi-RAT Coordinated Energy Saving
NR0SMRCSPS01	MRFD-160266	Multi-RAT Coordinated Symbol Power Saving (NR)	per Carrier per pRRU	Low-frequency TDD	Lamp Site gNodeB	No	Multi-RAT Coordinated Energy Saving
NR0SMRCCSD00	MRFD-170264	Multi-RAT Coordinated Channel Shutdown (NR)	per Cell	Low-frequency TDD	Macro gNodeB	No	Multi-RAT Coordinated Energy Saving
NR0S00RFDD00	MRFD-161263	RF Module Deep Dormancy (NR)	per RU	Low-frequency TDD High-frequency TDD	Macro gNodeB	No	Multi-RAT Coordinated Energy Saving

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0STNFDPS00	MRFD-161266	LTE TDD and NR Flash Dynamic Power Sharing (NR)	per Cell	Low-frequency TDD	MacrogNodeB	No	LTE and NR Power Allocation
NR0S0FNUSS00	MRFD-131263	LTE FDD and NR Uplink Spectrum Sharing (NR)	per Cell	Low-frequency TDD	MacrogNodeB	Yes	LTE FDD and NR SUL Dynamic Spectrum Sharing
NR0SFNSPCS01	MRFD-131263	LTE FDD and NR Uplink Spectrum Sharing (NR)	per Carrier per pRRU	Low-frequency TDD	Lamp Site gNodeB	Yes	LTE FDD and NR SUL Dynamic Spectrum Sharing

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S0LSBUE00	MRFD-171263	NSA Carrier Combination Selection Based on User Experience (NR)	per gNodeB	Low-frequency TDD High-frequency TDD	Macro gNodeB LampSite gNodeB	No	EPC-based NSA Experience Improvement
NR0S0NSBUE00	MRFD-171262	NSA/SA Selection Based on User Experience (NR)	per gNodeB	Low-frequency TDD	Macro gNodeB LampSite gNodeB	No	*NSA-SA Selection Based on User Experience* *EPC-based NSA Performance Enhancement*

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0S0BBUDD00	MRFD-181263	BBU Deep Dormancy (NR)	per gNodeB	Low-frequency TDD	Macro gNodeB Lamp Site gNodeB	No	Multi-RAT Coordinated Energy Saving
NR0S00CPES00	MRFD-191261	Intelligent Coordinated PSU Energy Saving (NR)	per gNodeB	Low-frequency TDD High-frequency TDD	Macro gNodeB	No	Green Site
NR0S00TPS000	MRFD-191262	Intelligent Peak Staggering (NR)	per gNodeB	Low-frequency TDD High-frequency TDD	Macro gNodeB	No	Green Site

Model	Feature ID	Feature Name	Sales Unit	RAT	NE	Is Cell Activation Affected by License Insufficiency	Mapping to FPD
NR0SRFMSDD00	MRFD-200261	RF Module Super Deep Dormancy (NR)	per RU	Low-frequency TDD High-frequency TDD	MacrogNodeB	No	Multi-RAT Coordinated Energy Saving
NR0SILIBAS00	MRFD-221264	iLiBattScan (NR)	per gNodeB	Low-frequency TDD High-frequency TDD	MacrogNodeB	No	Power Supply Management

Multimode feature license control items include all the feature license control items for the individual RATs of a multimode base station. That is, if an operator wants to use a feature on a multimode base station, the operator must purchase the feature license control items for each RAT of the multimode base station. If the feature license control item for any RAT is unavailable, this feature cannot take effect. For example, to use the multimode feature LTE TDD and NR Flash Dynamic Power Sharing, the license control item "LTE TDD and NR Flash Dynamic Power Sharing (LTE TDD)" on the LTE side and the license control item "LTE TDD and NR Flash Dynamic Power Sharing (NR)" on the NR side must be purchased. However, there are exceptions. For example, the Intelligent Peak Staggering feature, whose license information is listed in **Table 1-9**, can be used on a multimode base station as long as the corresponding license control item for one of the RATs involved is purchased.

Table 1-9 License control items required only in one RAT for a multimode base station

Feature Name	RAT-specific Feature ID and Name	Model
Intelligent Peak Staggering	GSM: MRFD-191202 Intelligent Peak Staggering (GSM)	LGB3IPS1
	UMTS: MRFD-191212 Intelligent Peak Staggering (UMTS)	LQW9IPS01
	LTE FDD: MRFD-191222 Intelligent Peak Staggering (LTE FDD)	LT1SNGPEST10
	LTE TDD: MRFD-191232 Intelligent Peak Staggering (LTE TDD)	LT4S0IPSTG00
	NR: MRFD-191262 Intelligent Peak Staggering (NR)	NR0S00TPS000